MATHS MANTHAN
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**Subject:** Mathematics **CBSE CLASS 10 – PERIODIC ASSESSMENT 10** **Time:** 60 Minutes
**Class:** Class X (CBSE) **Chapter 11: Areas Related to Circles** **MM: 30**

**Document Type:** Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

**CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:**

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

## SECTION A: MULTIPLE CHOICE MATHS MANTHAN | (6 × 1 = 6 Marks)

**Q1.** If the perimeter of a semicircular region is 72 cm, then its diameter is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 14 cm (B) 28 cm (C) 36 cm (D) 42 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

**Step 1: Formula and Radius Calculation** [0.5 Mark]

$$\text{Perimeter} = \pi r + 2r = r(\pi + 2) = 72 \implies r\left(\frac{22}{7} + 2\right) = 72$$

$$r\left(\frac{36}{7}\right) = 72 \implies r = \frac{72 \times 7}{36} = 14 \text{ cm}$$

**Step 2: Diameter Calculation** [0.5 Mark]

$$\text{Diameter } d = 2r = 2 \times 14 = 28 \text{ cm}$$

**Correct Option:** (B) 28 cm

Diameter of semicircular region = **28** cm

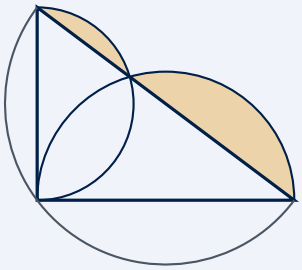
**Q2.** In a right triangle $ABC$ with legs $a$ and $b$, semicircles are drawn on $a, b$, and the hypotenuse $c$. The sum of the areas of the two crescents (lunes) outside the hypotenuse semicircle equals: [1]

(A) Area($\triangle ABC$) (B) $2 \times$ Area($\triangle ABC$) (C) $\frac{1}{2}$Area($\triangle ABC$) (D) $\pi \times$ Area($\triangle ABC$)

**▶ STEP-BY-STEP SOLUTION:** [1 Mark]

**Step 1: Theorem of the Lunes of Hippocrates** [0.5 Mark]

By Pythagoras' theorem: $a^2 + b^2 = c^2$.



$$\text{Sum of semicircles on legs} = \frac{1}{2}\pi\left(\frac{a}{2}\right)^2 + \frac{1}{2}\pi\left(\frac{b}{2}\right)^2 = \frac{\pi}{8}(a^2+b^2) = \frac{\pi}{8}c^2$$

which exactly equals the area of the semicircle on hypotenuse $c$! **Step 2: Deduction of Lunes Area** [0.5 Mark]

$$\text{Area(Lunes)} = \text{Area}(\triangle ABC) + \frac{\pi}{8}(a^2+b^2) - \frac{\pi}{8}c^2 = \text{Area}(\triangle ABC)$$

**Correct Option: (A) Area($\triangle ABC$)**

Sum of Lunes Areas = Area($\triangle$**ABC**)

**Q3.** A silver wire brooch in the form of a circle of diameter 35 mm has 5 diameters. The total length of wire used is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 285 mm (B) 110 mm (C) 175 mm (D) 385 mm

**▶ STEP-BY-STEP SOLUTION:** [1 Mark]

**Step 1: Wire Components** [0.5 Mark]

$$\text{Length} = \text{Circumference} + 5 \times \text{Diameter} = \pi d + 5d$$

**Step 2: Substitution** [0.5 Mark]

$$\text{Length} = \left(\frac{22}{7} \times 35\right) + (5 \times 35) = 110 + 175 = 285 \text{ mm}$$

**Correct Option: (A) 285 mm**

Total wire length = **285** mm

**Q4.** The area of a circle is 154 cm$^2$. The side of the largest square that can be inscribed in this circle is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) $7\sqrt{2}$ cm (B) 14 cm (C) 7 cm (D) $14\sqrt{2}$ cm

**▶ STEP-BY-STEP SOLUTION:** [1 Mark]

**Step 1: Radius and Diameter of Circle** [0.5 Mark]

$$\pi r^2 = 154 \implies \frac{22}{7}r^2 = 154 \implies r^2 = 49 \implies r = 7 \text{ cm} \implies d = 14 \text{ cm}$$

**Step 2: Side of Inscribed Square** [0.5 Mark]

$$\text{Diagonal } a\sqrt{2} = d = 14 \implies a = \frac{14}{\sqrt{2}} = 7\sqrt{2} \text{ cm}$$

**Correct Option: (A) $7\sqrt{2}$ cm**

Side of square $a = \mathbf{7\sqrt{2}}$ cm

**Q5.** The angle swept by the hour hand of a clock between 2:00 pm and 2:30 pm is: [1]

(A) 15° (B) 30° (C) 180° (D) 7.5°

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

**Step 1: Angular Speed of Hour Hand** [0.5 Mark]

In 60 minutes, the hour hand sweeps $\frac{360^\circ}{12} = 30^\circ$.

$$\text{Speed} = \frac{30^\circ}{60} = 0.5^\circ \text{ per minute}$$

**Step 2: Angle in 30 Minutes** [0.5 Mark]

$$\theta = 30 \times 0.5^\circ = 15^\circ$$

**Correct Option:** (A) 15°

Angle swept = **15°**

**Q6.** **Assertion (A):** The area of a sector of a circle of radius 6 cm with central angle 60° is $\frac{132}{7}$ cm$^2$.

**Reason (R):** The area of a sector of central angle $\theta$ (in degrees) and radius $r$ is given by Area = $\frac{\theta}{360^\circ} \times \pi r^2$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

**Step 1: Formula Verification** [0.5 Mark]

Reason (R) states the official sector area formula. Reason (R) is **True**.

**Step 2: Calculation Verification** [0.5 Mark]

$$\text{Area} = \frac{60^\circ}{360^\circ} \times \frac{22}{7} \times (6)^2 = \frac{1}{6} \times \frac{22}{7} \times 36 = \frac{132}{7} \text{ cm}^2$$

Assertion (A) is **True**, and Reason (R) correctly explains (A).

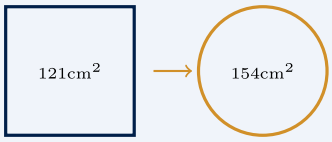
**Correct Option:** **(A) Both (A) and (R) are true and (R) is the correct explanation of (A).**

Correct Option: (**A**)

**Q7.** A steel wire bent in the form of a square encloses an area of 121 $cm^2$. If the same wire is bent into a circle, find the area enclosed by the circle. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

**Step 1: Perimeter from Square Area** [1 Mark]




$$\text{Area of square} = a^2 = 121 \implies a = 11 \text{ cm}$$

$$\text{Length of wire} = \text{Perimeter} = 4a = 4 \times 11 = 44 \text{ cm}$$

**Step 2: Area of Re-formed Circle** [1 Mark]

$$2\pi r = 44 \implies 2 \times \frac{22}{7} \times r = 44 \implies r = 7 \text{ cm}$$

$$\text{Area of circle} = \pi r^2 = \frac{22}{7} \times 7^2 = 154 \text{ cm}^2$$

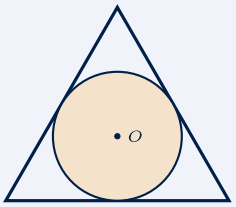
Area enclosed by circle = **154** $cm^2$

**Q8.** Find the area of the incircle of an equilateral triangle of side $14\sqrt{3}$ cm. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: Side $a = 14\sqrt{3}$ cm.

**Step 1: Compute Inradius $r$** [1 Mark]




$$r = \frac{a}{2\sqrt{3}} = \frac{14\sqrt{3}}{2\sqrt{3}} = 7 \text{ cm}$$

**Step 2: Area of Incircle** [1 Mark]

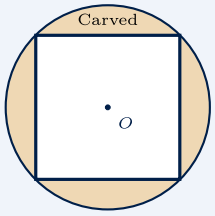
$$\text{Area} = \pi r^2 = \frac{22}{7} \times 7 \times 7 = 154 \text{ cm}^2$$

Area of incircle = **154** $cm^2$

**Q9.** A circular round table of radius 10 cm has a central square inscribed in it. The four circular segments outside the square are decorated with carvings. Find the total area of the four carved segments. (Use $\pi = 3.14$) [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Radius of circular table $r = 10$ cm.




**Step 1: Total Area of Circular Table** [1 Mark]

$$\text{Area of circle} = \pi r^2 = 3.14 \times (10)^2 = 314 \text{ cm}^2$$

**Step 2: Area of Inscribed Square** [1 Mark]

The diagonal of the inscribed square equals the circle diameter:

$$d = 2r = 2 \times 10 = 20 \text{ cm}$$

$$\text{Area of square} = \frac{1}{2}d^2 = \frac{1}{2} \times (20)^2 = 200 \text{ cm}^2$$

**Step 3: Area of Four Carved Segments** [1 Mark]

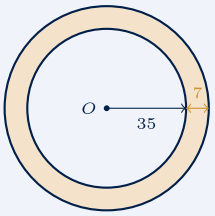
$$\text{Total Carved Area} = \text{Area of Circle} - \text{Area of Square} = 314 - 200 = 114 \text{ cm}^2$$

$$\boxed{\text{Total Area of 4 Carved Segments} = \mathbf{114} \text{ cm}^2}$$

**Q10.** The inner and outer diameters of a circular running track are 70 m and 84 m respectively. Find the cost of leveling the track at the rate of Rs. 5 per m$^2$. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Inner diameter = 70 m $\implies$ $r$ = 35 m. Outer diameter = 84 m $\implies$ $R = 42$ m.




**Step 1: Surface Area of Circular Track** [1.5 Marks]

$$\text{Area} = \pi(R^2 - r^2) = \frac{22}{7} \times (42^2 - 35^2) = \frac{22}{7} \times (42 - 35)(42 + 35)$$

$$= \frac{22}{7} \times 7 \times 77 = 22 \times 77 = 1694 \text{ m}^2$$

**Step 2: Total Leveling Cost** [1.5 Marks]

$$\text{Total Cost} = \text{Area} \times \text{Rate} = 1694 \times \text{Rs. } 5 = \text{Rs. } 8470$$

$$\boxed{\text{Track Area} = \mathbf{1694} \text{ m}^2, \quad \text{Total Cost} = \text{Rs. } \mathbf{8470}}$$

**Q11.** In a right triangle $ABC$ right-angled at $B$, $AB = 6$ cm and $BC = 8$ cm. Semicircles are described on $AB, BC$, and the hypotenuse $AC$ as diameters.

(a) Find the hypotenuse $AC$ and the area of $\triangle ABC$. **[1.5]**

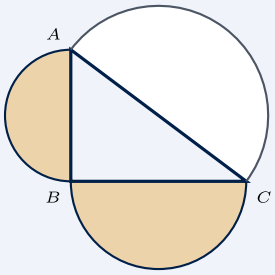
(b) Find the sum of the areas of the semicircles on $AB$ and $BC$. **[1.5]**

(c) Prove that the area of the two shaded crescents (lunes) formed outside the semicircle on $AC$ is exactly equal to the area of $\triangle ABC$. **[2]**

**► STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Given: Legs $AB = 6$ cm, $BC = 8$ cm, $\angle B = 90°$.

**Part (a): Hypotenuse $AC$ and $\triangle ABC$ Area** [1.5 Marks]




$$AC = \sqrt{AB^2 + BC^2} = \sqrt{6^2 + 8^2} = \sqrt{100} = 10 \text{ cm}$$

$$\text{Area}(\triangle ABC) = \frac{1}{2} \times 6 \times 8 = 24 \text{ cm}^2$$

**Part (b): Sum of Semicircles on Legs $AB$ and $BC$** [1.5 Marks]

Radii: $r_1 = 3$ cm, $r_2 = 4$ cm.

$$\text{Sum} = \frac{1}{2}\pi(3)^2 + \frac{1}{2}\pi(4)^2 = \frac{9\pi}{2} + \frac{16\pi}{2} = \frac{25\pi}{2} \text{ cm}^2$$

**Part (c): Proof for Lunes of Hippocrates** [2 Marks]

Area of semicircle on hypotenuse $AC$ ($r_3 = 5$ cm):

$$\text{Area(Semicircle } AC) = \frac{1}{2}\pi(5)^2 = \frac{25\pi}{2} \text{ cm}^2$$

$$\text{Total Area of Figure} = \text{Area}(\triangle ABC) + \text{Area(Semi } AB) + \text{Area(Semi } BC) = 24 + \frac{25\pi}{2}$$

$$\text{Area of Lunes} = \text{Total Area} - \text{Area(Semi } AC) = \left(24 + \frac{25\pi}{2}\right) - \frac{25\pi}{2} = 24 \text{ cm}^2$$

$$\text{Area(Lunes)} = \text{Area}(\triangle ABC) = 24 \text{ cm}^2 \quad \text{[Q.E.D.]}$$

(a) $AC = \mathbf{10}$ cm, Area($\triangle$) $= \mathbf{24}$ cm$^2$, (b) $\mathbf{\frac{25\pi}{2}}$ cm$^2$, (c) Lunes Area $= \mathbf{24}$ cm$^2$ = Area($\triangle$**ABC**)

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**Q12.** An equilateral triangle $ABC$ has side 12 cm. A circle is inscribed in it and another circle circumscribes it.

(a) Find the radius and area of the inscribed circle. **[1.5]**

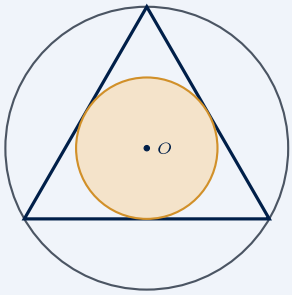
(b) Find the radius and area of the circumscribed circle. **[1.5]**

(c) Find the area of the circular ring between the circumscribed and inscribed circles. **[1]**

(d) Prove that the area of the circumscribed circle is exactly four times the area of the inscribed circle. (Use $\pi = 3.14$) **[1]**

**▶ STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Given: Side of equilateral triangle $a = 12$ cm.




**Part (a): Inscribed Circle ($r$)** [1.5 Marks]

$$r = \frac{a}{2\sqrt{3}} = \frac{12}{2\sqrt{3}} = 2\sqrt{3} \text{ cm}$$

$$\text{Area(Incircle)} = 3.14 \times (2\sqrt{3})^2 = 3.14 \times 12 = 37.68 \text{ cm}^2$$

**Part (b): Circumscribed Circle ($R$)** [1.5 Marks]

$$R = \frac{a}{\sqrt{3}} = \frac{12}{\sqrt{3}} = 4\sqrt{3} \text{ cm}$$

$$\text{Area(Circumcircle)} = 3.14 \times (4\sqrt{3})^2 = 3.14 \times 48 = 150.72 \text{ cm}^2$$

**Part (c): Area of Circular Ring** [1 Mark]

$$\text{Ring Area} = 150.72 - 37.68 = 113.04 \text{ cm}^2$$

**Part (d): Ratio Proof (4 : 1)** [1 Mark]

$$R = 2r \implies \text{Area(Circumcircle)} = \pi(2r)^2 = 4\pi r^2 = 4 \times \text{Area(Incircle)}$$

$$\frac{150.72}{37.68} = 4.0 \quad \text{[Proved]}$$

(a) $r = \mathbf{2\sqrt{3}}$ cm, 37.68 cm$^2$, (b) $R = \mathbf{4\sqrt{3}}$ cm, 150.72 cm$^2$, (c) Ring = **113.04** cm$^2$, (d) 4 : 1

## SECTION E: Case-Based / Applied Situation

MATHS MANTHAN | **(1 × 4 = 4 Marks)**

**Q13. Case Study: Eight-Ribbed Canopy Dome Umbrella Structural Geometry**

A canopy umbrella manufacturer constructs premium umbrellas featuring 8 equally spaced flexible ribs. In its fully open position, the umbrella fabric can be approximated as a flat planar circular disc of radius $r = 45$ cm.

(i) Determine the angle formed at the apex centre between any two consecutive ribs. **[1]**

(ii) Calculate the length of the curved rim arc connecting the tips of two consecutive ribs. **[1]**

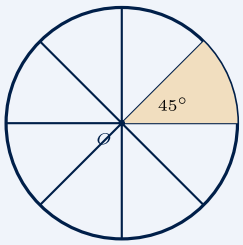
(iii) Calculate the fabric surface area contained between two consecutive ribs. $\left(\text{Use } \pi = \frac{22}{7}\right)$ **[2]**

**— OR —**

Calculate the total circular fabric area of the open canopy. **[2]**

**▶ STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Given: Radius $r = 45$ cm, Number of equally spaced ribs $n = 8$.




**Part (i): Central Angle between Two Consecutive Ribs** [1 Mark]

$$\theta = \frac{360^\circ}{8} = 45^\circ$$

**Part (ii): Length of Curved Rim Arc** [1 Mark]

$$l = \frac{45^\circ}{360^\circ} \times 2\pi r = \frac{1}{8} \times 2 \times \frac{22}{7} \times 45 = \frac{990}{28} = \frac{495}{14} \approx 35.36 \text{ cm}$$

**Part (iii): Fabric Area between Two Consecutive Ribs** [2 Marks]

$$\text{Area} = \frac{1}{8} \times \pi r^2 = \frac{1}{8} \times \frac{22}{7} \times (45)^2 = \frac{1}{8} \times \frac{22}{7} \times 2025 = \frac{22275}{28} \text{ cm}^2 \approx 795.54 \text{ cm}^2$$

**— OR (Alternative Choice) —**

**Total Circular Fabric Area of Canopy:** [2 Marks]

$$\text{Total Area} = \pi r^2 = \frac{22}{7} \times (45)^2 = \frac{22}{7} \times 2025 = \frac{44550}{7} \text{ cm}^2 \approx 6364.29 \text{ cm}^2$$

(i) $\theta = \mathbf{45^\circ}$, (ii) Arc $l = \mathbf{\frac{495}{14}}$ cm $\approx \mathbf{35.36}$ cm, (iii) Sector Area $= \mathbf{795.54}$ cm$^2$ (OR Total $= \mathbf{6364.29}$ cm$^2$

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